FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSF 3253 :

INTELLIGENT SYSTEM GROUP ASSIGNMENT

TITLE:

HYBRID INTELLIGENCE IN AVIATION: OPTIMIZING FLIGHT CONTROL USING FUZZY EVOLUTIONARY NETWORKS

GROUP 12

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1.0 Introduction

The aviation sector are changing quickly. In recent years, the flight systems has shifted from human control to more automated and advanced technologies. Therefore, modern aircraft now rely on flight control systems to maintain a stable direction, and safety flight. These systems are widely used in Commercial aircraft, military drones, rescue helicopters, and unmanned aerial vehicles (UAVs). The primary purpose of flight control systems is to ensure safe and stable operation during different flight conditions.

However, because the flight environment is unpredictable, controlling an aircraft is a difficult task. Weather disruptions, turbulence, abrupt wind changes, and other challenging-to-model external forces are commonplace for aircraft. Control decisions may also be impacted by noisy or marginally inaccurate data produced by flight sensors. Particularly during crucial stages like takeoff, landing, and emergency maneuvers, these uncertainties make flight control extremely difficult.

In the past, linear mathematical models like proportional-integral-derivative (PID) controllers have been used to design aircraft flight control systems. Under steady and predictable circumstances, these controllers operate effectively. Real-world flight dynamics, however, are frequently non-linear, which means that minor alterations to the surrounding environment can have significant and unanticipated effects on aircraft behavior. Because of this, conventional control techniques might find it difficult to react appropriately when flight conditions become unstable or change quickly.

Researchers are increasingly investigating Artificial Intelligence (AI) methods for flight control applications in order to overcome these constraints. Fuzzy Evolutionary Networks are one useful strategy. By representing inputs using linguistic terms like "low," "medium," and "high," fuzzy logic allows the system to deal with uncertainty in a manner akin to human reasoning. In the meantime, evolutionary genetic algorithms use system performance over time to automatically optimize the control rules.

The goal of this project is to use a fuzzy evolutionary network to create a hybrid intelligent flight control system. The suggested system seeks to enhance aircraft stability, adaptability, and safety in unpredictable and non-linear flight environments by fusing fuzzy logic with evolutionary optimization.

2.0 Problem Background & Problem Solution

2.1 Problem Background

Designing a reliable autopilot system for aircraft, especially unstable platforms such as helicopters and Unmanned Aerial Vehicles (UAVs), remains a challenging task in aerospace engineering. Traditional control approaches often struggle to handle the complex nature of flight dynamics. There are three main issues that limit the effectiveness of conventional flight control systems.

Non-Linearity and Instability:
Many aircraft need constant control modifications to sustain steady flight because they are inherently unstable. For instance, without continuous input corrections, a helicopter cannot stay in the air. Because of the extreme non-linearity of airplane dynamics, a slight alteration in wind direction or speed can result in a significant and unanticipated shift in the aircraft's motion. Under such circumstances, traditional linear controllers frequently fail to react promptly and precisely.

Uncertainty and Noise:
Sensor data is used by flight control systems to make judgments, but noise and uncertainty frequently impact this data. Environmental factors like turbulence can't always be accurately represented, and sensor values can change. Conventional control systems demand precise numerical values, while human pilots can use experience to understand ambiguous conditions like "high turbulence". This limitation can lead to inappropriate or unsafe control actions when sensor data is slightly inaccurate.

Lack of Interpretability:
Flight control systems rely on sensor data to make decisions, but noise and uncertainty often affect this data. Sensor values can fluctuate, and environmental elements like turbulence can't always be adequately reflected. While human pilots can utilize expertise to comprehend

ambiguous conditions like "high turbulence," conventional control systems require exact numerical values.

2.2 Problem Solution

This research suggests using a fuzzy evolutionary network for aircraft flight control in order to overcome the drawbacks of conventional control techniques. This method substitutes an intelligent and adaptable control framework for inflexible mathematical models.

The system's central decision-making component is fuzzy logic. Fuzzy Inference Systems (FIS) use human-like reasoning to translate ambiguous or uncertain inputs, like aircraft speed or turbulence level, into meaningful control actions. This enables the system to make realistic and seamless control modifications.

The fuzzy system is optimized via the evolutionary algorithm component. A Genetic Algorithm (GA) automatically enhances the rule base through simulation and assessment, as opposed to manually creating and fine-tuning fuzzy rules. Effective control rules are kept throughout several generations, while ineffective ones are eliminated.

Because of this, the suggested controller can manage noise and uncertainty, handle non-linear flight behavior, and make transparent decisions using interpretable "If–Then" fuzzy rules.

3.0 Methodology (algorithm, framework/model, and justification)

3.1 Components of Fuzzy Evolutionary Network

Fuzzy Evolutionary Network combines evolutionary optimization and fuzzy inference to produce an interpretable, adaptive autopilot that manages nonlinear dynamics, sensor uncertainty, and environmental disturbances. Therefore, it is important to know the basic components before going through the methodology. Below are the main key components to address the problem statement;

- **Fuzzy Inference System (FIS)**: Takagi‑Sugeno‑Kang (TSK) or Mamdani controller that maps sensor inputs to actuator commands using human‑readable If–Then rules.
- **Evolutionary Optimizer**: Genetic Algorithm (GA), Differential Evolution (DE), or Particle Swarm Optimization (PSO) that evolves FIS parameters and structure.
- **Simulation and Validation Environment**: High‑fidelity nonlinear aircraft model, hardware‑in‑the‑loop (HIL) rig, and staged flight test setup.
- **Fitness and Safety Module**: Multi‑objective evaluation that enforces performance, robustness, and safety constraints.

3.2 Fuzzy Evolutionary Network's Algorithm

Pack membership function centers and widths, rule on/off flags, and TSK consequent coefficients into a real or mixed vector by encoding the controller as a single chromosome. Decrypt each genome into a FIS, start a population with randomly selected and expert-seeded individuals, then assess it in a nonlinear simulator or HIL under representative conditions like normal flight, gusts, sensor noise, and mass or center-of-gravity alterations. Evaluate candidates using a multi-objective fitness that enforces strict safety penalties for actuator limitations and stability violations, minimizes tracking error and control effort, maximizes robustness across scenarios, and penalizes complexity. Using evolutionary operators to select, crossover, and mutate for GA or mutation and recombination for DE to retain elites. Then, repeat the process using parallel simulation until convergence or budget constraints are met, archiving the Pareto front and optimizing the best controllers.

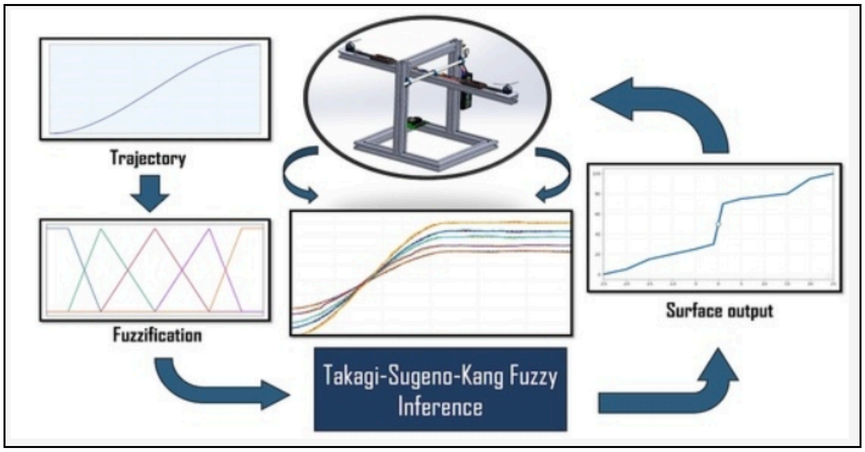

Figure 3.1: Graphical abstract of TSK controller

The internal operation of a Takagi–Sugeno–Kang (TSK) fuzzy controller is shown in figure 3.1. Fuzzification is used to transform crisp inputs into fuzzy membership values once a reference trajectory is first supplied as input. The TSK inference system evaluates these fuzzy values and produces weighted outputs using a number of rules. The physical system or plant (illustrated in the center) is driven by the smooth surface output created by combining the results. This loop illustrates the real-time mapping of inputs to control actions by a single fuzzy controller.

3.3 Framework and justification

The control model used in this methodology is a Takagi–Sugeno–Kang (TSK) fuzzy inference system because its linear consequent functions produce smooth, continuous actuator commands and can be effectively optimized. The several fuzzy inference techniques used for flight-control optimization are shown in Figure 3.2. Particle Swarm Optimization is a lightweight substitute for low-dimensional tuning issues, Genetic Algorithms are employed when discrete rule selection is necessary, and Differential Evolution is chosen for effective continuous parameter optimization. High-fidelity six-degree-of-freedom (6-DOF) simulations, hardware-in-the-loop testing, and incremental flight tests are the first steps in the tiered process used to validate the system. After validation, unnecessary rules are eliminated, and local optimization techniques are used to improve the resulting parameters. This approach uses fuzzy reasoning to handle nonlinear system dynamics and sensor noise, maintains human-interpretable If-Then rules to satisfy certification requirements, and takes advantage of global evolutionary optimization to generate reliable and broadly applicable flight-control solutions.

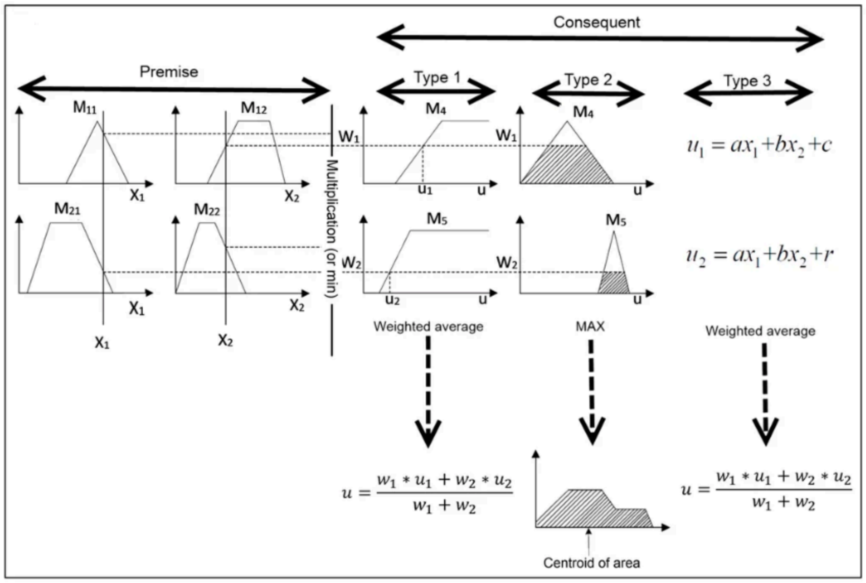

Figure 3.2: Type of Fuzzy Inference

Figure 3.2 illustrates the various fuzzy system types described above using a two-rule, two-input fuzzy inference system. The output function of the popular Mamdani type fuzzy system, type 2, is based on the total fuzzy output; some of these include area centroid, maximum of maxima, minimum of maxima, etc. The Takagi-Sugeno type fuzzy system is type 3.

4.0 Result and Discussion (expected results, benefits, advantages, and impact)

4.1 Expected Results: Case Study Analysis

Using a simulated velocity control case study at Guarulhos International Airport in São Paulo, the effectiveness of the suggested Fuzzy Evolutionary Network was assessed. Due to stringent safety and speed regulations, the simulation depicts an aircraft making the transition from cruise flight to landing, which is one of the most crucial stages of flight.

Throughout the descending procedure, the system showed steady and seamless velocity control. During cruise flight, the aircraft's speed was roughly 800 knots; during approach, it was 250 knots; and while landing, it was 80 knots. By removing sudden or step-like control actions, the application of Gaussian membership functions helped to smooth throttle adjustments. As a result, the velocity profile closely followed the target speed curve while complying with airport safety constraints.

The effectiveness of the fuzzy rule foundation was examined in addition to control performance. To accommodate for different flight situations, the fuzzy system first produced a vast number of rules. Redundant and inefficient rules were automatically eliminated following the application of the Genetic Algorithm optimization. This preserved consistent control performance while drastically reducing the size of the rule basis. A more compact and effective controller was produced by the optimized rule set, making it appropriate for real-time implementation in onboard embedded aviation systems.

4.2 Benefits and Advantages

Safety and Robustness:
One of the main advantages of the proposed system is improved safety. The fuzzy logic component is naturally tolerant to uncertainty and sensor noise. When sensor readings are slightly inaccurate due to turbulence or weather disturbances, the system continues to produce stable control outputs instead of reacting aggressively. This reduces the risk of oscillations or loss of control that may occur in traditional controllers.

Adaptability:
Control performance was analyzed together with the fuzzy rule foundation's effectiveness. The fuzzy system initially generated a large number of rules to account for various flight conditions. After the Genetic Algorithm optimization was applied, redundant and ineffective rules were automatically removed. This significantly reduced the size of the rule basis while maintaining constant control performance. The optimized rule set resulted in a more efficient and compact controller that is suitable for real-time deployment in onboard embedded aircraft systems.

Human-Like Decision Making:
The system can make decisions that are comparable to those of a skilled human pilot thanks to the fuzzy inference method. It strikes a balance between several goals, including preserving safety, guaranteeing passenger comfort, and increasing fuel economy. Compared to traditional black-box control techniques, this human-like logic makes the system more accessible and understandable.

4.3 Impact

There is a lot of potential for the suggested Fuzzy Evolutionary Network, especially in the area of unmanned aerial vehicles (UAVs). It makes it possible for drones to function independently in difficult and dangerous situations where human assistance may be restricted or dangerous, such as disaster areas, forest fires, and extreme weather.

This technology is a step toward next-generation autopilot systems in commercial aviation. Such technologies can serve as intelligent partners that assist pilots by adjusting to unpredictable circumstances and improving overall flight safety, as opposed to serving as passive control tools. Future aviation operations could be safer, more dependable, and more effective with the use of hybrid intelligent control systems.

5.0 Conclusion & Recommendation

Modern flight control systems can be effectively optimized by the application of hybrid intelligence in aviation, which combines evolutionary genetic algorithms with fuzzy logic. Non-linear dynamics, sensor noise, and unpredictable climatic conditions are among the difficulties faced by aircraft, particularly unstable platforms like helicopters and unmanned aerial vehicles. Adaptive and intelligent control is required in these situations since traditional linear controllers frequently fail to maintain stability.

These issues are addressed by the suggested Fuzzy Evolutionary Network, which combines adaptive optimization with human-like reasoning. Fuzzy logic generates smooth and accurate control operations by interpreting imprecise or ambiguous flight data, such as turbulence levels, speed deviations, or altitude variations. In the meantime, the evolutionary method reduces redundancy and increases computational efficiency by automatically optimizing the fuzzy rules. The case study results showed that while optimizing its internal rule base for real-time implementation, the system could successfully maintain smooth velocity control throughout cruise-to-landing transitions.

Several suggestions for further research are made in light of these findings. Real-time flight simulations should be used to evaluate the system in a range of scenarios, such as extreme turbulence, wind shear, or partial system failures. Multi-objective optimization to balance passenger comfort, fuel economy, and flight safety could also be investigated. Adaptive neuro-fuzzy systems may also enable ongoing learning while in flight, enhancing performance and robustness even more.

In summary, the use of fuzzy evolutionary networks in hybrid intelligence offers a clever and useful way to address the difficulties of contemporary flight control. This technology can enable the future of autonomous and semi-autonomous aviation systems by enhancing the safety, dependability, and flexibility of both human and unmanned aircraft by tackling non-linearity, uncertainty, and instability.

6.0 Figures and Illustrations

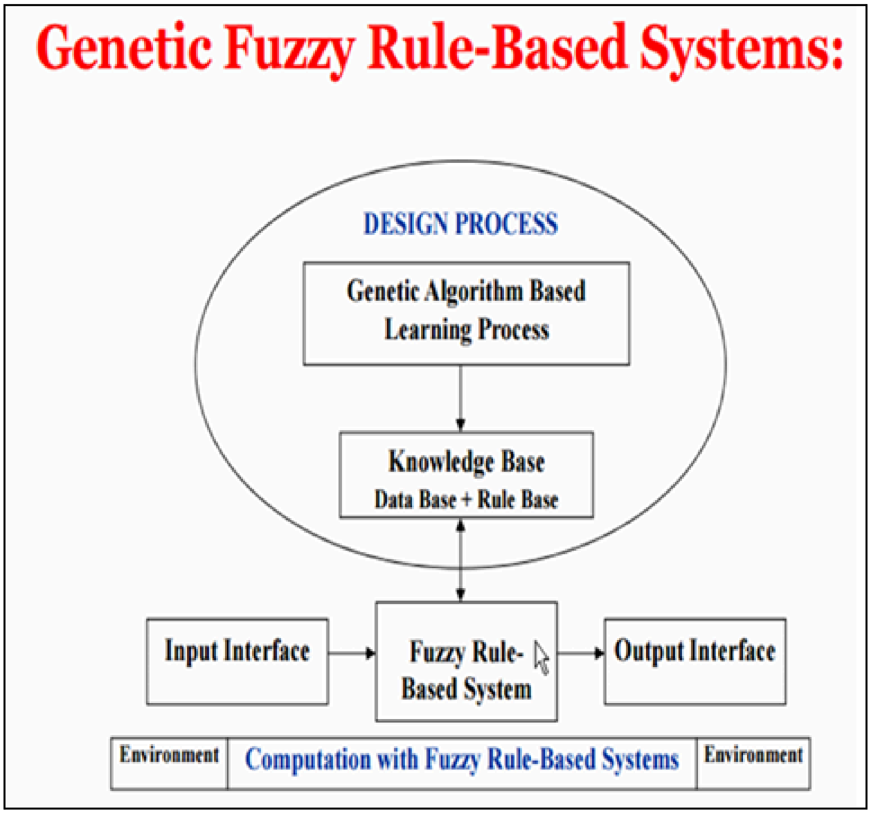

Figure 6.1: Genetic Fuzzy Rule Based System

The typical techniques for combining fuzzy logic with genetic algorithms GA-Optimized Fuzzy Systems and Fuzzy-Driven GA Adaptation. of order to increase searching efficiency, fuzzy-driven GA adaptation uses fuzzy logic controllers that can dynamically determine the parameters of genetic algorithms, such as crossover probability or mutation probability. Applying Genetic Algorithms as an optimizing algorithm to solve searching issues in fuzzy systems, such as optimizing membership functions or rules, is known as GA-Optimized Fuzzy Systems.

The combination of Global Search (GA) and Local Search (LS) to overcome individual restrictions is one of hybridization's primary advantages. Strength-wise, the LS locates local optima with excellent accuracy, while the GA quickly identifies promising locations. GA's slowness in arriving at the ideal answer is a drawback, while LS's tendency to become stuck is a weakness.

The Premature Convergence Paradox, Representation, and Complexity are the main problems and obstacles in hybridization. The Premature Convergence Paradox describes how fuzzy logic controllers can assist in achieving a more balanced approach to exploration and exploitation, but they are plagued by premature convergences, which are related to crossover operators and parameter adaption. Aside from that, the difficulties are representation and complexity, which relate to halting criteria and genotype complexity.

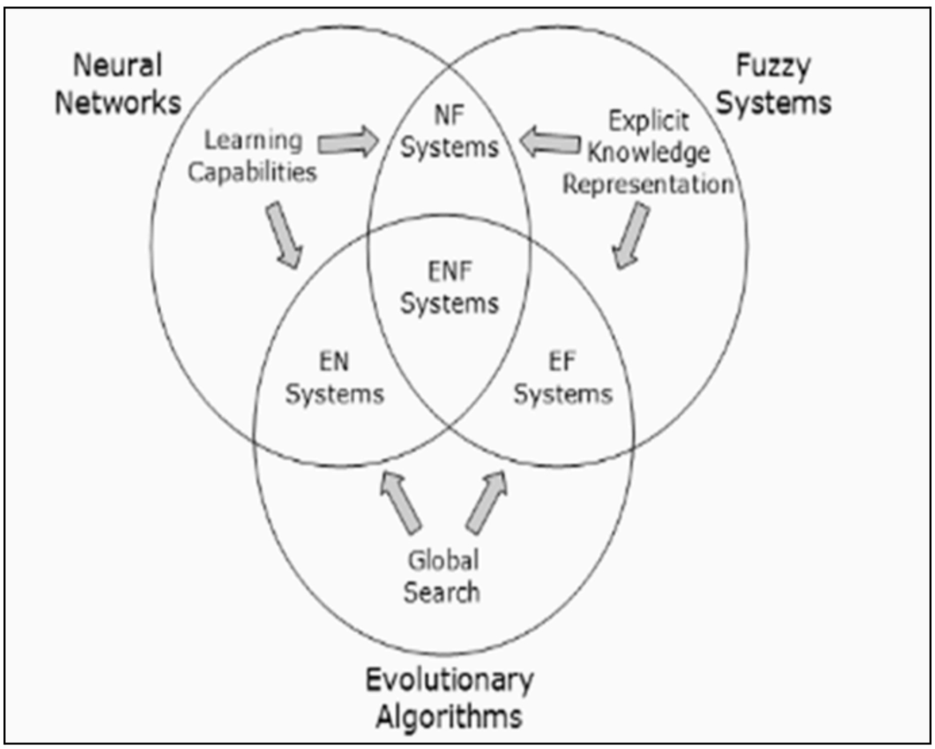

Figure 6.2: Evolutionary Neuro Fuzzy Model

The applications that combine the learning capabilities of neural networks and evolutionary algorithms are described in Figure 6.2. the main benefit of separate "Soft Computing" pillars that work together to overcome each specific flaw. In specifics, the Fuzzy System offers an approximation of human decision-making logic, but it lacks the capacity for learning. The multi-layer architecture offered by the neural network is capable of learning patterns but is challenging to decipher. By functioning as a global optimization, the Evolutionary Algorithms improve the system's parameters. The system can overcome the limitations of fuzzy logic systems that can learn from available data built from fuzzy inference systems that are multi-layered in the neural architecture by integrating the primary value of separate "Soft Computing" pillars.

Fuzzy systems software computer paradigms include both the intelligence component and the adaptive process.The system's capacity to adapt while remaining fit amid shifting conditions is known as the adaptation process. Environmental adjustment and structural alteration make up the adaption process. The linguistic variables are altered by the environmental adjustment. By altering the components of the surrounding surroundings, the structural alteration makes the fuzzy inference system more suitable.

The intelligence components describe how the system responds to and picks up on novel circumstances. Using a neural network to alter and apply knowledge to contexts is an example of abstract thinking. By measuring performance against predetermined objectives, the system is able to improve outputs based on objective success. The system uses an adaptive database framework that supports highly sophisticated decision-making systems (IF/THEN rules) and applies evolutionary algorithms.

7.0 References (APA citation style)

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8.0 Video Link Presentation

Link Youtube : https://youtu.be/1TvJ4f6wI1U